

Retrieving yearly forest growth from satellite data: FORMS-T, a deep learning based approach

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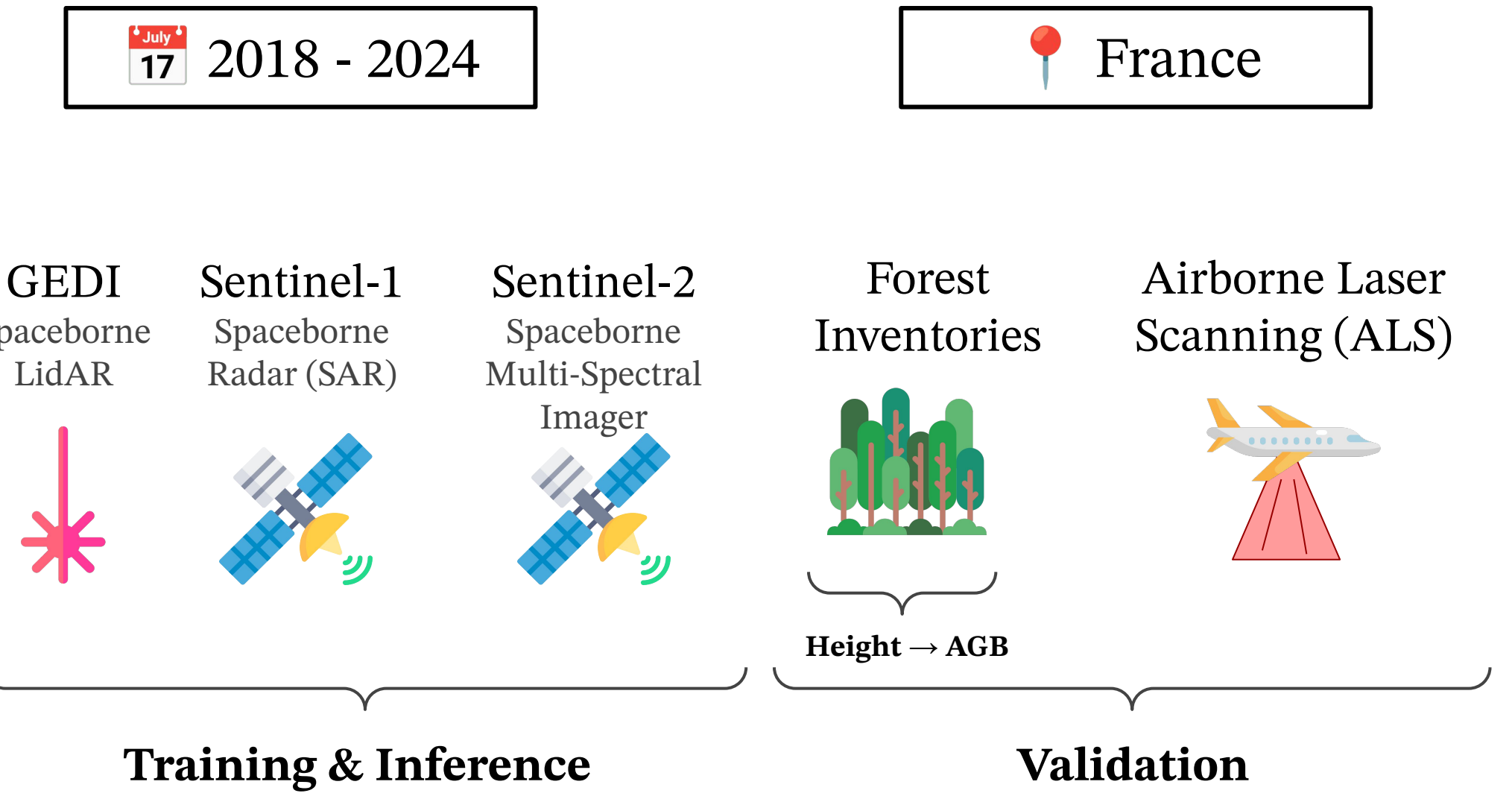
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The need for growth monitoring

Most satellite-based forest monitoring focuses on detecting **losses** (e.g. deforestation, disturbance), but **growth** is equally important for understanding forest carbon dynamics, ecosystem recovery, and sustainable management.

Data sources

The **GEDI Lidar data** give **sparse height measurements** over the study area, whereas **Sentinel-1 and Sentinel-2** give **continuous images**. **Forest inventories** (*in-situ* measurements) are used in the calibration of **allometric equations** to derive aboveground biomass and standing wood volume density maps.



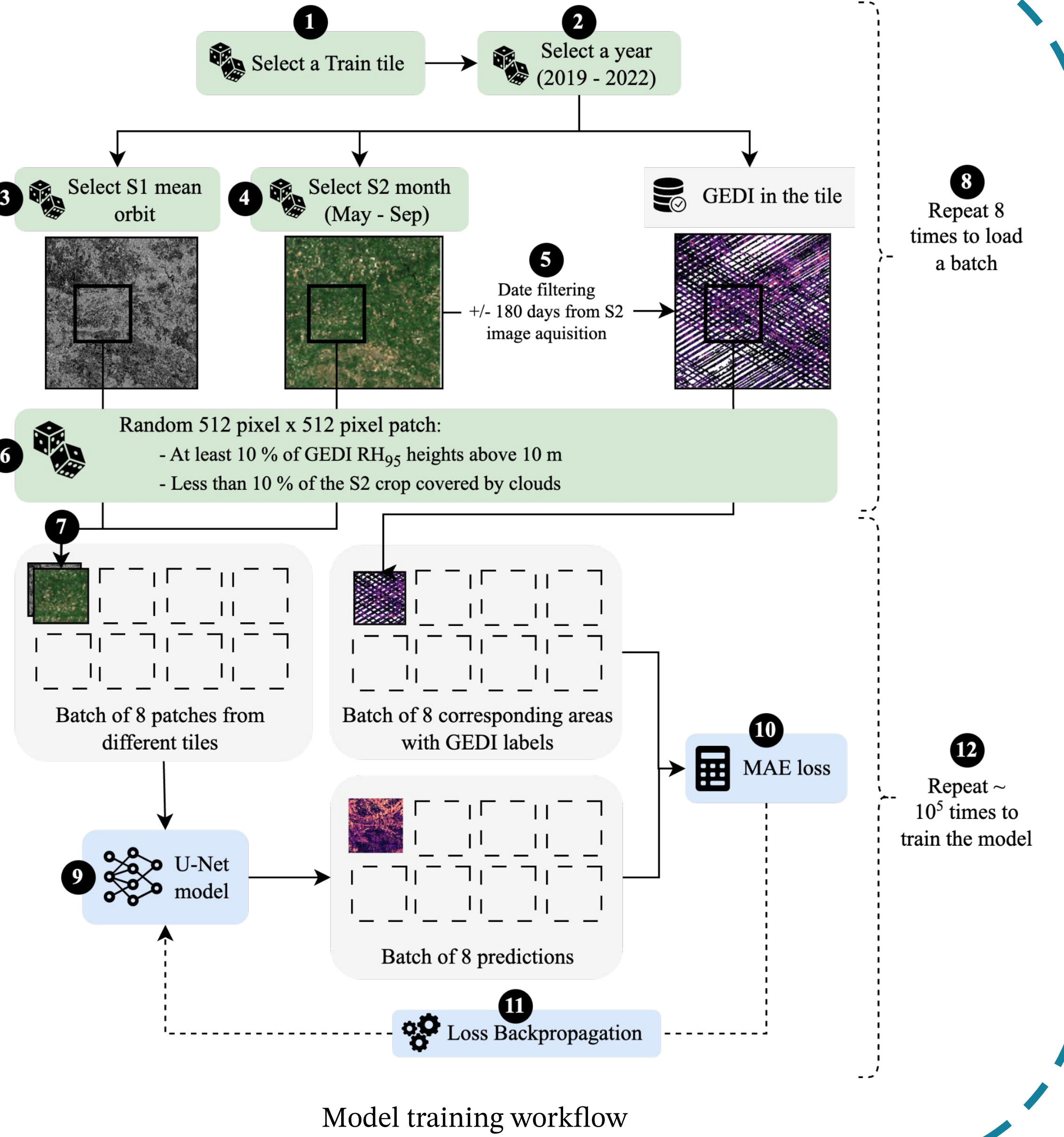
Per-Image Training: A Year-Agnostic Model

Unlike most approaches that rely on annual or seasonal composites, FORMS-T is trained on **individual Sentinel-2 scenes**. This exposes the model to a wide range of acquisition conditions (phenology, cloud cover, atmospheric variability), allowing it to learn **invariant features** related to forest structure. As a result, the model is **not tied to a specific year or season**, and can generate consistent predictions across multiple years, making it truly **year-agnostic**.

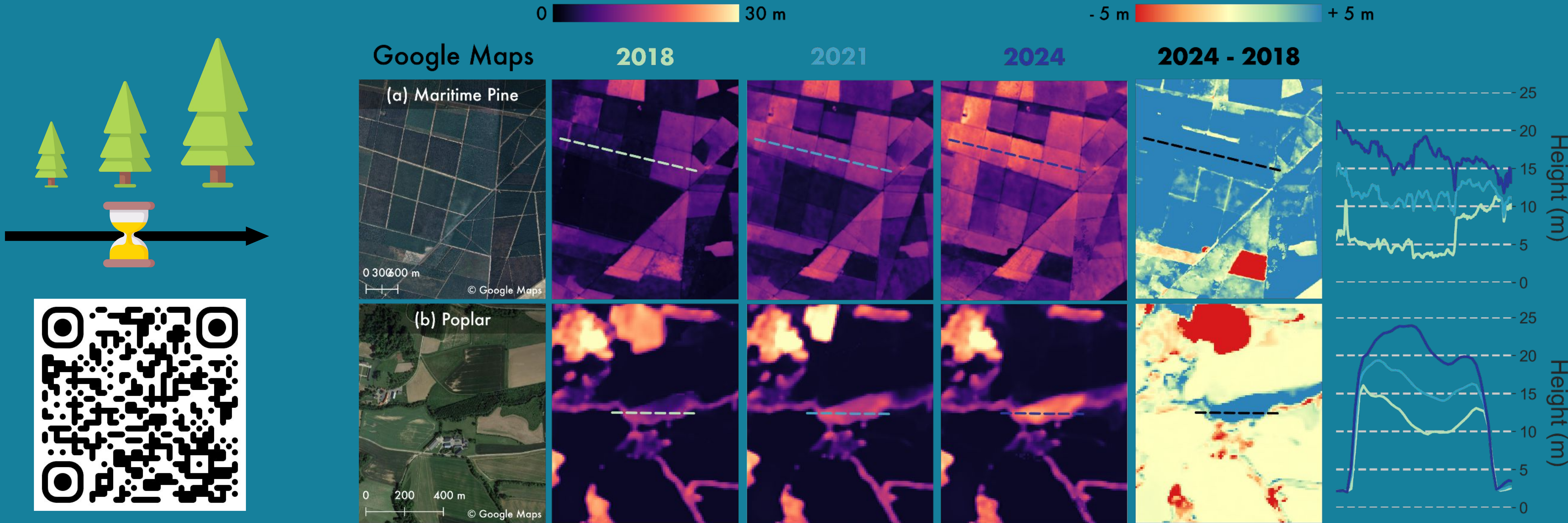
Post-Processing: Trend-Aware Smoothing

To enhance temporal consistency, we apply a **Mann-Kendall trend test** to each pixel's time series. Based on whether a significant trend is detected, we apply **either a 3-year or 5-year moving average**. This simple yet effective method reduces noise while preserving real growth signals, improving the reliability of forest monitoring.

The FORMS-T Framework



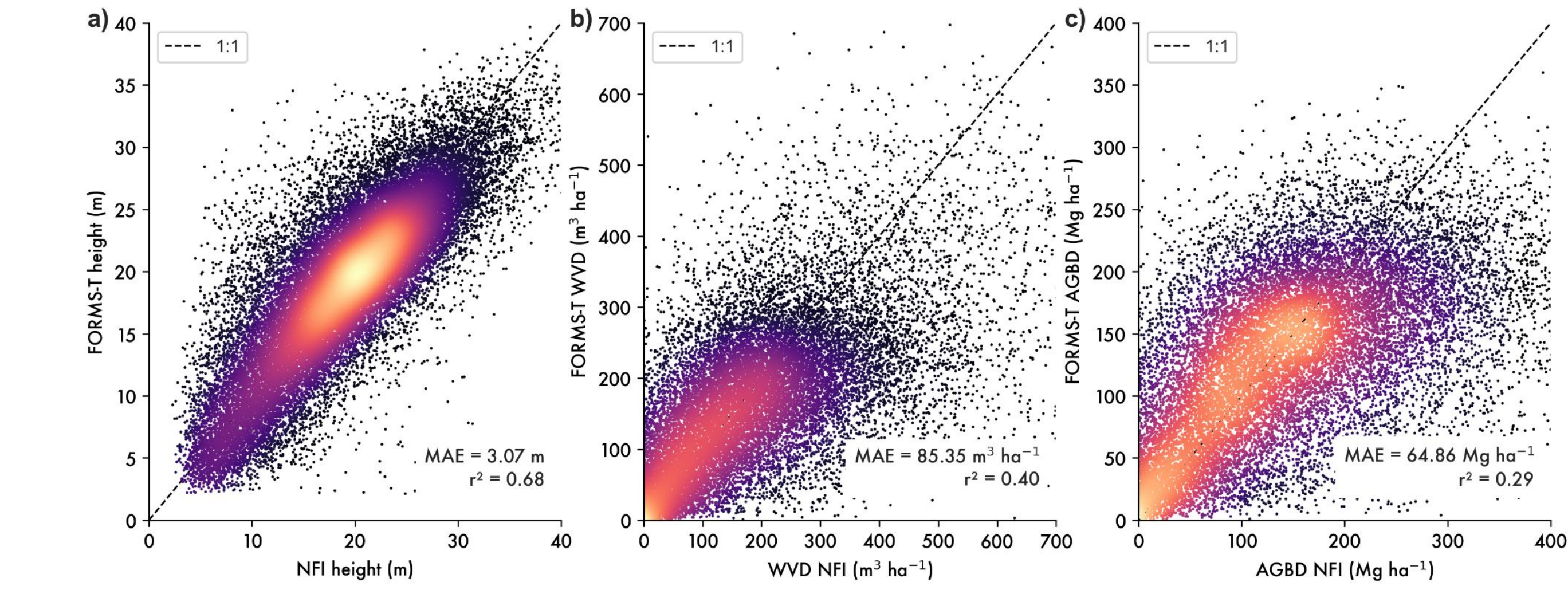
We can capture annual forest growth from satellite Data



Results

Validation with NFI dataset

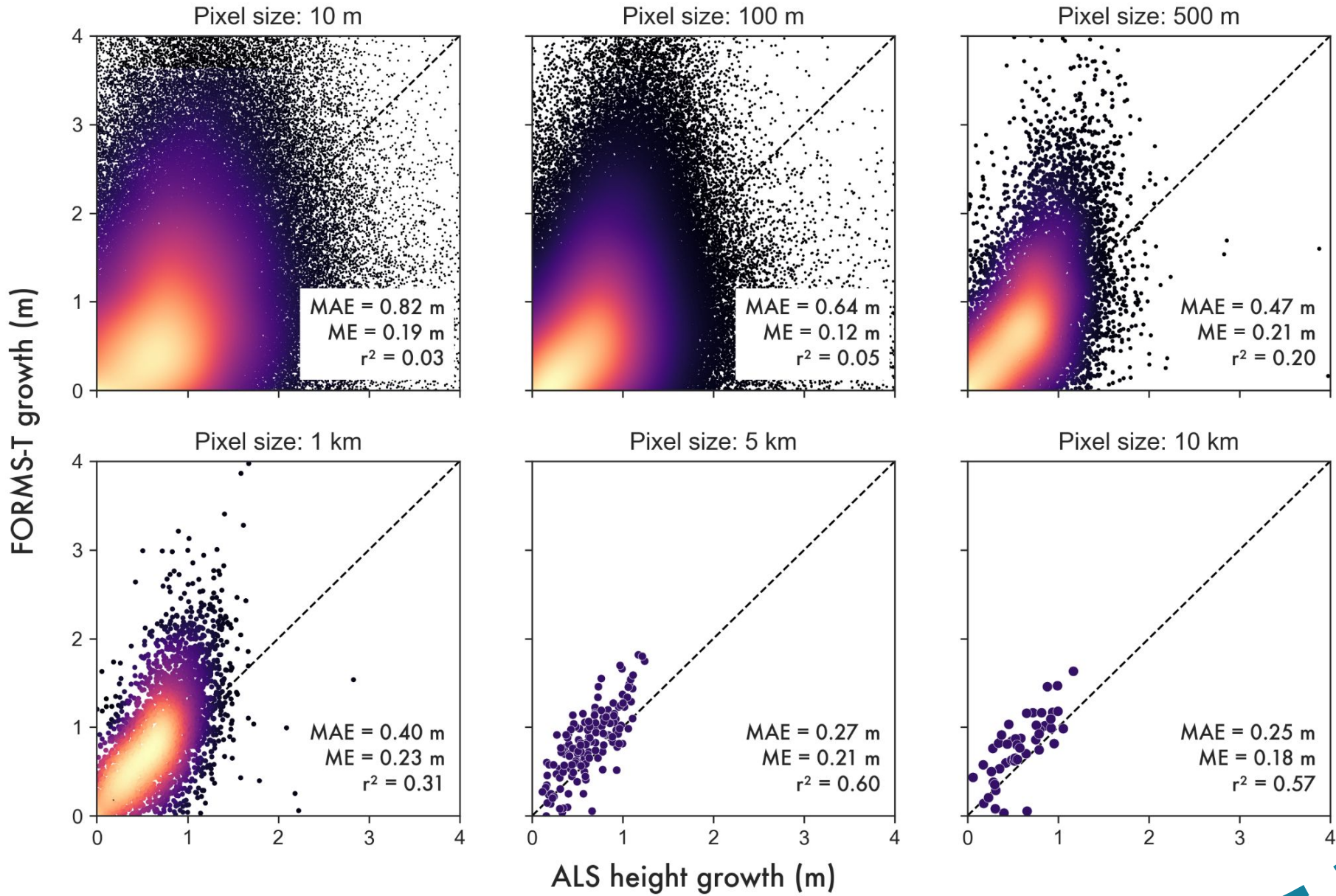
FORMS-T canopy height predictions were validated against NFI field measurements across all years from 2018 to 2024, yielding a **mean absolute error (MAE) of 3.07m**. Performance varies by forest type and terrain: **flat, homogeneous regions** (e.g., Landes) show lower errors, while mountainous or structurally complex areas show higher variability.



Capturing forest growth from space

A key achievement of FORMS-T is its ability to detect **forest height growth over time** using satellite data. While individual-year predictions carry uncertainty, **temporal aggregation** reveals consistent growth signals at local scales (~5km). This is confirmed through validation with **successive airborne LiDAR (ALS) campaigns**, showing that FORMS-T can track **incremental canopy height change**, a crucial step toward monitoring forest carbon uptake and ecosystem recovery.

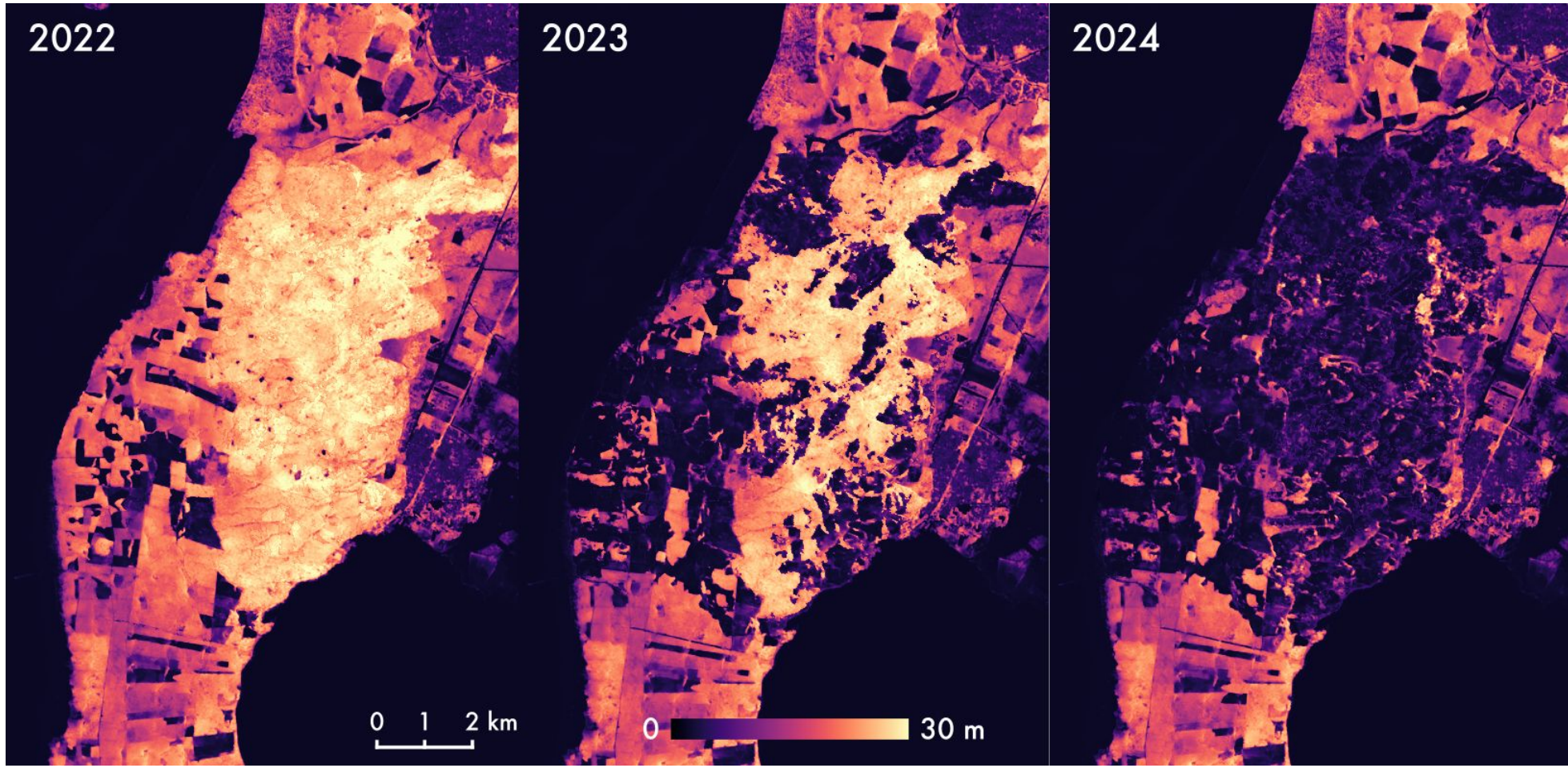
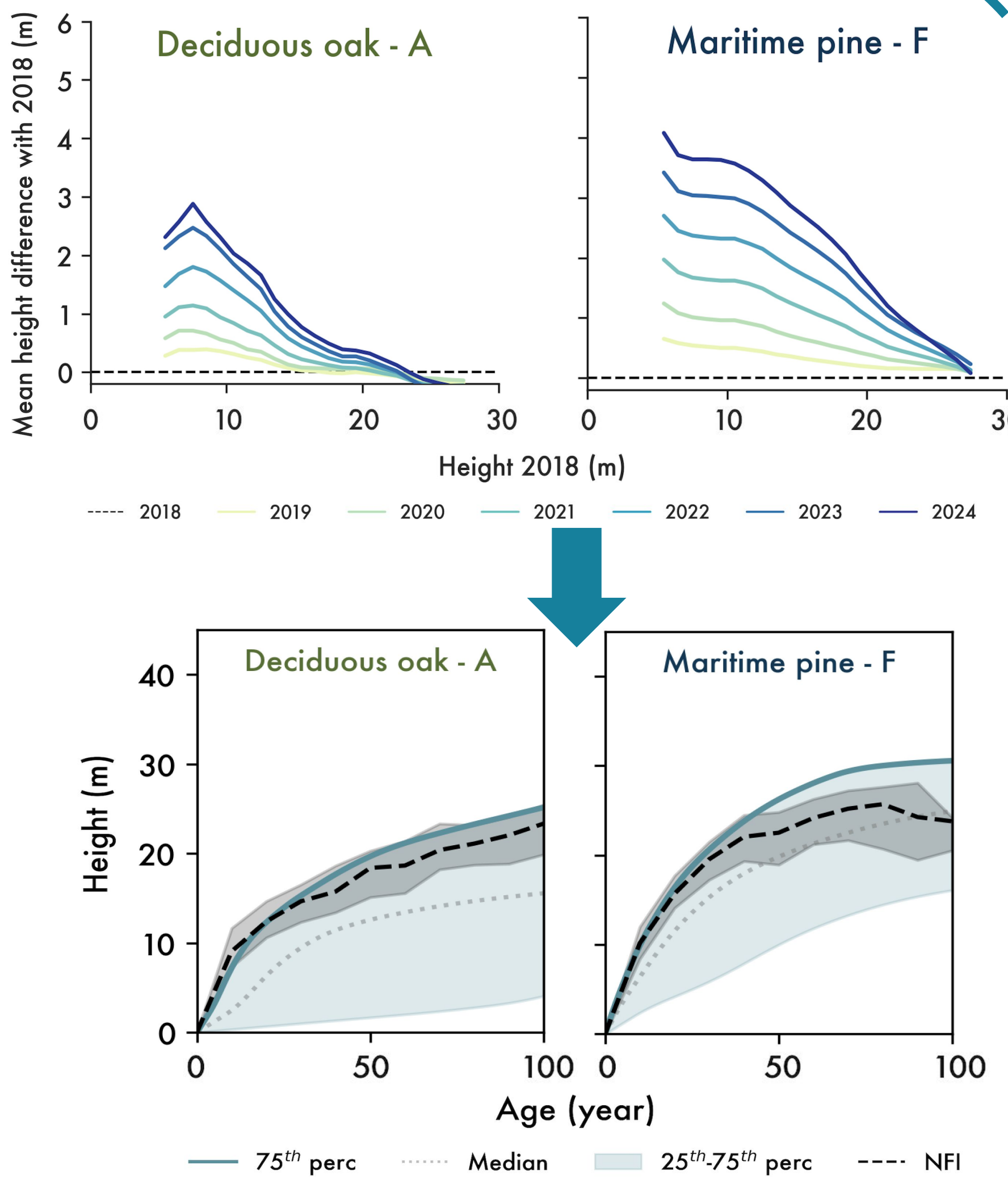
- Comparison of ALS growth for 10-15 m height classes in the Vosges mountains (France) vs FORMS-T growth for the same period (2020-2022).
- Aggregating results at 5 km show that FORMS-T can capture a consistent growth signal.



Applications

Applications: From Growth Curves to Carbon Monitoring

FORMS-T supports a wide range of forest monitoring applications. By aggregating height data over time and by species, we can derive **species-specific growth curves** directly from satellite observations. Combined with forest loss detection, FORMS-T enables the **tracking of both carbon gains (growth) and losses (disturbance, harvest)**. These insights are valuable for **carbon stock assessments, forest management planning, and long-term ecosystem monitoring**, supporting national and European climate and biodiversity goals.



Ability to monitor forest losses: FORMS-T canopy height in the "La Teste de Buch" forest (44.57° N, -1.18° E) in 2022, 2023 and 2024. A huge fire occurred in this area in July 2022.

Limitations and Sources of Error

Despite its strengths, FORMS-T has several limitations. The **mean absolute error (~3m)** makes it challenging to detect subtle height changes at the pixel level. The model performs best in **flat, homogeneous forests**, and less reliably in **mountainous or structurally complex areas**. Temporal accuracy is also influenced by **cloud cover distribution**, which varies the timing of observations. Finally, **uncertainty in GEDI reference data and model generalization limits** may affect performance in underrepresented forest types.